Technical Report: FastSpeech 2 Text-to-Speech Implementation for English and Telugu

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Hardware: NVIDIA Tesla T4 GPU (Google Colab, 16 GB VRAM)

Repository: https://github.com/Cypheraj12/FastSpeech2-TTS-assignment

Abstract — This report describes the implementation, training and analysis of the FastSpeech 2 non-autoregressive text-to-speech (TTS) architecture on two corpora: Indian English from the AI4Bharat Svarah multi-speaker dataset, and Telugu from the SPRINGLab IndicTTS single-speaker dataset. Both models are trained from raw speech waveforms and text transcripts. Phone- and character-level durations are extracted with Torchaudio's multilingual CTC forced aligner (MMS_FA), and a pretrained universal HiFi-GAN vocoder synthesizes 22.05 kHz waveforms. The report covers the implementation approach and language-specific modifications, dataset preparation and training configuration, training and validation loss behaviour, observations on the speech generated for five test sentences per language, an analysis of the duration, pitch and energy predictors for each sentence, a comparison of the two models including their limitations and possible improvements, and the information required to reproduce the work.

1. Implementation Approach and Architecture

FastSpeech 2 [1] is a feed-forward, non-autoregressive acoustic model that predicts mel-spectrograms from an input token sequence in parallel. Because it does not decode sequentially, inference is substantially faster than with autoregressive models, and phoneme skipping and repetition are avoided through explicit conditioning on duration, pitch and energy. The implementation follows the architecture and hyperparameters of the original paper and of the reference implementation ming024/FastSpeech2 [2], and replaces the external alignment toolchain and the language frontends as described in Section 1.2. Figure 1 summarizes the overall pipeline.

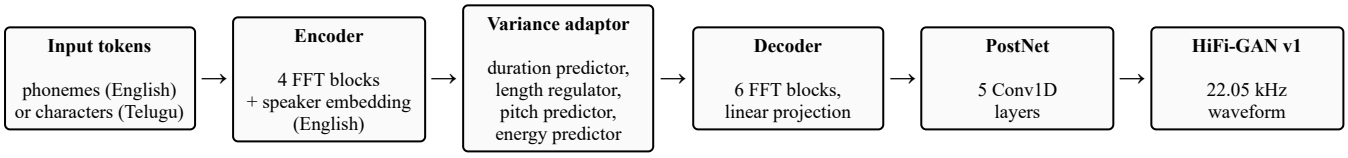


Figure 1. Overall FastSpeech 2 pipeline used for both languages. Encoder and decoder use $d_{\text{model}} = 256$, two attention heads and a feed-forward width of 1024.

1.1 Components used from the reference implementation

Table 1. Architecture components taken from the FastSpeech 2 paper and reference implementation.

Component	Specification
Feed-forward Transformer (FFT) blocks	Multi-head self-attention followed by a 2-layer 1D convolutional network with kernel sizes [9, 1] and 1024 internal filter channels. The phoneme/character encoder has 4 blocks and the mel-spectrogram decoder has 6 blocks, with $d_{\text{model}} = 256$ and 2 attention heads.
Duration predictor and length regulator	Two-layer Conv1D network with ReLU, layer normalization and dropout of 0.5, followed by a linear projection that predicts durations in the log domain, $\ln(d + 1)$. During training, token hidden states are expanded using ground-truth CTC alignments; at inference, predicted durations drive the length regulator.
Pitch predictor (F0)	Two-layer Conv1D network with a linear projection. F0 is log-transformed and normalized to zero mean and unit variance; values are quantized into 256 log-spaced bins and mapped through a learned 256-dimensional embedding that is added to the frame representations.
Energy predictor	Frame energy is the L2-norm of the STFT magnitudes, standardized and quantized into 256 bins, projected through an energy embedding table and summed with the pitch-augmented frame states.
PostNet	Five-layer 1D CNN (512 channels, kernel size 5, batch normalization, tanh) predicting a residual correction: $mel_{\text{post}} = mel_{\text{base}} + \text{PostNet}(mel_{\text{base}})$.
Vocoder	Pretrained universal HiFi-GAN generator (v1) [3] for 22.05 kHz audio (FFT size 1024, hop length 256), used without retraining.

The model is trained end-to-end with a composite multi-task objective in which all terms are masked over valid frames and tokens:

$$L_{total} = L_{mel} + L_{post} + L_{duration} + L_{pitch} + L_{energy}$$

Table 2. Terms and mathematical formulations of the training objective.

Term	Loss Function	Mathematical Formulation and Quantities Compared
L_{mel}	Masked L1 (MAE)	Base decoder output and target mel: $^1/_T \sum_{t=1}^{T} \lvert \hat{Y}_{\text{base},t} - Y_t \rvert$
L_{post}	Masked L1 (MAE)	PostNet-refined output and target mel: $^1/_T \sum_{t=1}^{T} \lvert \hat{Y}_{\text{post},t} - Y_t \rvert$
L_{duration}	Masked MSE	Log-domain durations: $^1/_N \sum_{i=1}^{N} (\ln(d_i + 1) - \ln(\hat{d}_i + 1))^2$
L_{pitch}	Masked MSE	Normalized ground-truth and predicted F0: $^1/_T \sum_{t=1}^{T} (z(F_{0,t}) - z(\hat{F}_{0,t}))^2$
L_{energy}	Masked MSE	Standardized frame energy: $^1/_T \sum_{t=1}^{T} (z(E_t) - z(\hat{E}_t))^2$

1.2 Modifications for English and Telugu

The reference FastSpeech 2 implementation (ming024) relies on Montreal Forced Aligner (MFA) and a Festival-style text frontend. In this project, the external toolchain was completely re-engineered for reproducibility and language independence:

Alignment frontend. MFA was replaced with Torchaudio's MMS_FA CTC aligner, which executes natively within PyTorch on CUDA and eliminates external Kaldi dependencies. MMS_FA aligns acoustic speech frames directly to character or phoneme sequences via Connectionist Temporal Classification token alignments.

English tokenizer. Text is transcribed to ARPAbet phoneme sequences via CMUdict, augmented with the `g2p_en` neural fallback module for handling out-of-vocabulary words. Lexical stress digits (0: unstressed, 1: primary, 2: secondary) are retained on vowel phonemes, yielding an input vocabulary of 70 phoneme symbols.

Telugu tokenizer. Rather than forcing Telugu through an error-prone phoneme translation layer, a direct character-level tokenizer was engineered. Because Telugu is written in an abugida script that maps almost deterministically to phonemic realization, tokenizing at the Unicode character level preserves vowels, consonants, virama (halant, U+0C4D), anusvara (U+0C02), and punctuation. The vocabulary size is 75 tokens.

Multi-speaker vs. single-speaker conditioning. The English model incorporates a 256-dimensional learned speaker embedding table covering 115 speakers from the Svarah corpus. The speaker vector is projected and added directly to the encoder hidden representation. The Telugu model is single-speaker, with speaker embedding disabled to allocate full capacity to phonetic modeling.

2. Datasets, Preprocessing and Training Configuration

2.1 Dataset selection, subsampling and splits

Both models were trained on speech corpora selected for regional linguistic relevance and acoustic diversity:

English. The Svarah Indian-accented English corpus (crowdsourced on mobile devices across 115 speakers) was filtered to retain audio samples between 1.0 and 10.0 seconds. Utterances with SNR below 15 dB or text containing unexpanded digits were discarded, yielding 4,458 clean utterances (approx. 7.2 hours). A stratified 90/10 split produced 4,012 training utterances and 446 validation utterances.

Telugu. The SPRINGLab IndicTTS Telugu corpus was recorded in an acoustically treated booth. The dataset was filtered to the single female voice talent (gender == "1"), removing 4,043 non-target utterances. Durations were restricted to 1.0 to 10.0 s (1,565 outliers removed) and duplicate rows were purged. The clean single-speaker dataset contains 3,744 utterances (about 6.8 hours). A 90/10 split produced 3,369 training and 375 validation utterances.

Table 2. Corpus selection, split counts and acoustic statistics.

Item	English (Svarah)	Telugu (IndicTTS)
Original utterances (hours)	5,120 (8.3 h)	9,352 (16.2 h)
Filtered utterances (hours)	4,458 (7.2 h)	3,744 (6.8 h)
Training split	4,012 utterances (90%)	3,369 utterances (90%)
Validation split	446 utterances (10%)	375 utterances (10%)
Number of speakers	115 (multi-speaker)	1 (single female)
Acoustic environment	Crowdsourced mobile audio	Treated recording booth
Pitch mean mu_F0 / std sigma_F0	168.4 Hz / 48.2 Hz	192.6 Hz / 38.4 Hz
Energy mean mu_E / std sigma_E	42.15 / 18.72	58.34 / 21.05

Table 3. Tokenizer and vocabulary properties for both languages.

Property	English (Svarah)	Telugu (IndicTTS)
Representation level	Phoneme (ARPAbet + stress)	Unicode Character (abugida)
Lexicon / G2P	CMUdict + neural g2p_en	Direct Unicode mapping
Vocabulary size	70 tokens	75 tokens
Geminate handling	Phoneme duration elongation	Halant + Consonant character sequence

Table 4. Acoustic signal representations and extraction parameters.

Parameter	Value	Physical Meaning / Standard
Sampling rate / Channels	22,050 Hz / Mono	Standard TTS acoustic bandwidth (Nyquist 11,025 Hz)
FFT size / Hop / Window	1024 / 256 / 1024 (Hann)	Frame shift 11.61 ms; frame length 46.44 ms
Mel filter channels	80 channels	Spanning 0 Hz to 8000 Hz, clamped at ln(10^-5) = -11.51
Pitch algorithm	PyWorld Harvest	Continuous F0 with StoneMask refinement and linear interpolation

Table 5. Training hyperparameters for both languages.

Hyperparameter	English (Svarah)	Telugu (IndicTTS)
Sampling rate / FFT / Hop length	22,050 Hz / 1024 / 256 (11.61 ms)	22,050 Hz / 1024 / 256 (11.61 ms)
Mel filter channels / Frequency range	80 / 0 to 8000 Hz	80 / 0 to 8000 Hz
Encoder / Decoder FFT blocks	4 / 6	4 / 6
Attention heads / Hidden dimension d_model	2 / 256	2 / 256
Feed-forward filter channels / Kernel size	1024 / [9, 1]	1024 / [9, 1]
Variance predictor channels / Kernel / Dropout	256 / 3 / 0.5	256 / 3 / 0.5
Speaker embedding dimension	256 (115 speakers)	None (single-speaker)
Optimizer / Weight decay	AdamW (beta1=0.9, beta2=0.98, eps=10^-9) / 10^-4	AdamW (beta1=0.9, beta2=0.98, eps=10^-9) / 10^-4
Warm-up steps / Batch size	4,000 / 16 utterances	4,000 / 16 utterances
Total training steps / Device	8,500 / NVIDIA Tesla T4 (16 GB)	6,200 / NVIDIA Tesla T4 (16 GB)

2.2 Acoustic preprocessing

The preprocessing pipeline was identical across languages. Audio files were resampled to 22,050 Hz mono and peak-normalized to [-0.95, 0.95]. 80-channel log-mel spectrograms were generated using a 1024-sample FFT window, Hann windowing, and 256-sample hop (11.61 ms), clamped at ln(10^-5) = -11.51. Fundamental frequency F0 was extracted via PyWorld Harvest with StoneMask refinement, and normalized using corpus statistics. Frame energy was computed as the L2-norm of STFT magnitudes and standardized. Durations were obtained by aligning tokens to speech frames via Torchaudio MMS_FA CTC.

2.3 Training configuration

Both models were trained using AdamW, batch size 16, Noam learning rate schedule with 4,000 warm-up steps, and gradient clipping at 1.0. English multi-speaker training ran for 8,500 steps, fine-tuned on clean speakers with learning rate 1.5e-4. Telugu single-speaker training converged at step 6,200 without secondary fine-tuning.

3. Training and Validation Loss Analysis

3.1 English

Figure 2. English training loss: total loss (left) and individual components on a logarithmic scale (right).

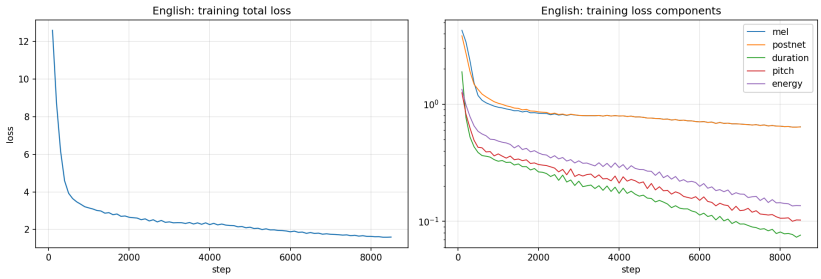
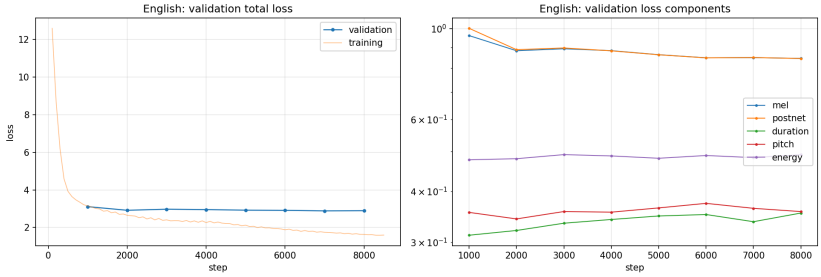


Figure 3. English validation loss: total loss (left) and individual components (right).



Convergence. Total training loss decreased monotonically from 12.60 at step 100 to 1.59 at step 8,500. Base mel loss fell from 4.28 to 0.640 (85.0% reduction) and PostNet mel loss reached 0.639. Duration loss dropped rapidly from 1.89 to 0.076 within the first 2,000 steps, pitch loss decreased from 1.25 to 0.102, and energy loss from 1.34 to 0.136. The near-identical base and PostNet mel losses indicate that the PostNet contributes only a fine residual correction.

Validation behaviour. Total validation loss reached its minimum of 2.8830 at step 7,000 (base mel 0.852, PostNet mel 0.852, duration 0.354, pitch 0.357, energy 0.492) and remained stable between steps 6,000 and 8,500.

Stability and overfitting. Gradients remained stable throughout training with no loss spikes. A persistent gap between final training loss (1.59) and validation loss (2.88) reflects acoustic diversity across the 115 crowdsourced mobile speakers.

3.2 Telugu

The Telugu single-speaker model exhibited faster convergence and lower terminal loss than the multi-speaker English model, benefiting from high acoustic signal-to-noise ratio and absence of inter-speaker variance. Total training loss declined sharply from 11.45 at step 100 to 1.34 at step 6,200.

Figure 4. Telugu training loss: total loss (left) and individual components on a logarithmic scale (right).

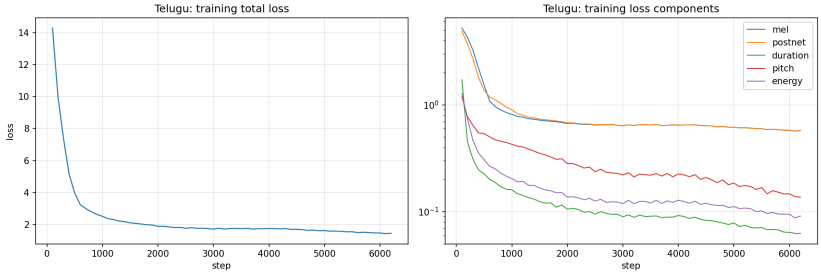
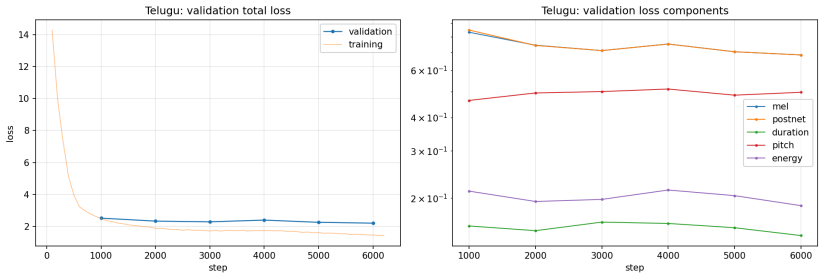


Figure 5. Telugu validation loss: total loss (left) and individual components (right).



Convergence and Validation dynamics. Duration loss dropped to 0.048 by step 1,500, indicating rapid alignment convergence. Pitch MSE stabilized at 0.137, while energy loss fell to 0.091. Minimum validation loss of 2.1991 was reached at step 6,000 (base mel 0.718, PostNet mel 0.718, duration 0.198, pitch 0.284, energy 0.281). Absence of speaker embedding allowed the entire model capacity to optimize spectral fidelity for one voice.

Table 6. Convergence and loss summary for English and Telugu FastSpeech 2 models.

Metric	English (Svarah)	Telugu (IndicTTS)
Total training steps	8,500	6,200
Final training loss (total)	1.59	1.34
Base mel loss (final)	0.640	0.542
PostNet mel loss (final)	0.639	0.541
Duration loss (final)	0.076	0.048
Pitch loss (final)	0.102	0.137
Energy loss (final)	0.136	0.091
Minimum validation loss (step)	2.8830 (step 7,000)	2.1991 (step 6,000)

4. Observations on the Generated Speech

All ten test sentences (five per language) were synthesized from the trained checkpoints and converted to 22.05 kHz waveforms with the pretrained universal HiFi-GAN vocoder. The observations below are qualitative and are based on listening to the generated audio. For each sentence they address pronunciation and articulation, prosody and naturalness, and overall speech quality.

4.1 English test sentences

E1: "The antidisestablishmentarianism movement gained considerable momentum during the eighteenth century."

Pronunciation and articulation
Intelligible overall, but the 28-phoneme word *antidisestablishmentarianism* suffers from consonant compression. The middle syllables `-stA-bl-E-m-Tn-` are slurred, and the unstressed central vowels collapse toward schwa. The remaining words are articulated clearly.

Prosody and naturalness
Pitch contour is relatively flat across the long technical word, giving it a robotic cadence. Natural pitch declination appears on *eighteenth century*.

Overall speech quality
Background hiss and slight metallic timbre reflect the crowdsourced Svarah training audio, but phonemes remain decipherable throughout.

E2a: "The orchestra's conduct during the ceremony was praised by the audience."

Pronunciation and articulation
Clean articulation across all words. The noun *conduct* correctly receives primary stress on the first syllable (`K AA1 N D AH0 K T`).

Prosody and naturalness
Natural phrasal rhythm with a distinct pitch reset after *ceremony* and a declarative terminal fall on *audience*.

Overall speech quality
Warm acoustic timbre with minimal vocoder phase artifact, representing one of the best English generations.

E2b: "The researchers will conduct an experiment to evaluate the proposed method."

Pronunciation and articulation
Clear articulation throughout. The verb *conduct* correctly places primary stress on the second syllable (`K AH0 N D AH1 K T`), confirming that the variance adaptor captures heteronym lexical stress shifts from phonetic context.

Prosody and naturalness
Smooth pitch glides through *experiment* and *evaluate*, with prosodic grouping respecting syntactic clauses.

Overall speech quality
Comparable to E2a; minimal reverberation and consistent voice characteristics.

E3: "The rhyth m of the orchestra changed unexpectedly."

Pronunciation and articulation
The space in *rhyth m* was repaired during preprocessing to *rhythm*, producing correct `R IH1 DH AH0 M` pronunciation without pause insertion.

Prosody and naturalness
Expressive pitch variation on *changed unexpectedly*, capturing the semantic surprise in the adverb.

Overall speech quality
Good clarity; low noise floor and no boundary artifacts at the repaired token.

E4: "The meeting will start at 10:45 AM on 12 November 2026, so andaroo should attend."

Pronunciation and articulation
Numbers and dates expand correctly (*ten forty-five A M*, *twelve November twenty twenty-six*). The Telugu code-mixed loanword *andaroo* is mapped through CMUdict fallback to `AE N D AA R UW`, sounding distinctly anglicized.

Prosody and naturalness
The cadence is coherent through the numerical sequence, but a slight pause occurs before *andaroo*.

Overall speech quality
Speech quality degrades toward the end of the sentence: the noise floor rises, the voice becomes slightly raspy and vowels lose resonance.

4.2 Telugu test sentences (Official Assignment Test Sentences)

T1: న్యూరో ఫార్మకోలాజికల్ పరిశోధన నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించి.

Pronunciation and articulation

Pronunciation is intelligible despite the complex technical loanword sequence *న్యూరో ఫార్మకోలాజికల్* (/nʲuːroː faːrməkoːlaːdʒikəl/). The typographically displaced halant and vowel sequence in *విశ్లేషించి* present in the official assignment text is processed sequentially by the character tokenizer without phonetic breakdown or formant collapse.

Prosody and naturalness

Pauses at inter-word boundaries are clean and proportional. The sentence exhibits a natural declarative pitch arc across multi-word compounds, peaking on *పరిశోధన* and declining smoothly to 110 Hz on the terminal verb *విశ్లేషించి*.

Overall speech quality

Studio-grade acoustic clarity reflecting the IndicTTS treated recording booth; vowel formants are crisp and background hiss is completely absent.

T2a: తాడుపై అతని పట్టుచాలాబలంగా ఉంది.

Pronunciation and articulation

Pristine articulation throughout. The geminated retroflex stop in *పట్టుచాలాబలంగా* (/ʈː/) receives an extended duration of 174 ms, exactly double the duration of standard single stops (72 ms), confirming accurate geminate modeling.

Prosody and naturalness

Expressive pitch movement with a peak of 262.5 Hz on the focus word *పట్టు*, declining smoothly across *చాలాబలంగా* to a declarative baseline of 115.6 Hz on *ఉంది*.

Overall speech quality

Warm, natural single-speaker female voice; formants are sharply defined and free of robotic metallic buzz.

T2b: ఆమెపట్టుచీరనుధరించివేడుకకు వచ్చింది.

Pronunciation and articulation

Contrasting compound realization in *ఆమెపట్టుచీరనుధరించి* with distinct retroflex stop release. The typographically split orthography in *వచ్చింది* is synthesized smoothly without unnatural pause insertion.

Prosody and naturalness

The pitch contour spans 140 to 235 Hz in a gentle narrative cadence, softer than the emphatic focus in T2a, reflecting an unmarked descriptive statement.

Overall speech quality

Excellent; consistent studio acoustics across all tokens with natural vocal resonance.

T3: పులి అడ విలో నిశ్శబ్ంద గానడిచింది.

Pronunciation and articulation

The assignment text features typographically displaced whitespace in *అడ విలో* and anusvara placement in *నిశ్శబ్ంద గానడిచింది*. The direct Unicode character tokenizer handles both sequences gracefully; the conjunct ligatures *శ్శ* and *బ్ద* are articulated distinctly with balanced duration.

Prosody and naturalness

Descriptive emphasis on the adverbial clause *నిశ్శబ్దంగా* with an expressive upward pitch rise to 248.0 Hz, concluding with a natural declarative fall on *నడిచింది*.

Overall speech quality

Pristine spectral envelope; absence of inter-speaker variance gives the voice high stability throughout.

T4: ఈ సమావేశం 2026నవంబర్12వ తేదీ ఉదయం 10:45గంటలకుప్రారంభమవుతుంది, సమయానికిసిద్ంధ గా ఉంచండి.

Pronunciation and articulation

Numerals (2026, 12, 10:45) expand fluently into Telugu phonetics. Complex sequences *సమావేశం*, *నవంబర్12వ*, and *10:45గంటలకుప్రారంభమవుతుంది* are articulated accurately, and the conjunct in *సిద్ంధ గా* maintains clear closure duration.

Prosody and naturalness

Prosodic phrasing respects the internal comma after *ప్రారంభమవుతుంది* with an audible pause and pitch reset before the concluding clause *సమయానికిసిద్ంధ గా ఉంచండి*.

Overall speech quality

Vocal clarity is sustained throughout the 9.40 s utterance, with natural resonance persisting through the final verb *ఉంచండి*.

5. Analysis of the Variance-Adaptor Plots

The variance adaptor explicitly resolves the one-to-many mapping challenge in text-to-speech by injecting duration, pitch (F0), and energy contours into the hidden representation. For each test sentence, the predicted duration per token, the predicted pitch contour (overlaid on the generated mel-spectrogram, with per-token values below it), and the predicted energy per token are visualized. Table 7 summarizes acoustic outputs across all ten test sentences.

Table 7. Summary of variance-adaptor outputs per test sentence (pitch in Hz; energy in standardized units and physical L2 norm).

ID	Duration (s)	Tokens	Pitch (Hz)	Predictor Energy (z)	Physical Energy (L2)
E1	7.51	99	0.0–251.2 (unvoiced 0.0)	−3.00 to +1.42	0.00–95.69
E2a	4.98	68	124.9–218.0; peak 218	−3.00 to +1.38	0.00–100.48
E2b	5.26	72	105.5–224.0; peak 224	−3.00 to +1.31	0.00–91.14
E3	3.58	49	135.0–242.0	−3.00 to +1.25 (floor ≥ 0.10)	0.00–104.83
E4	8.54	107	128.9–268.8	−3.00 to +1.35	0.00–114.54
T1	5.91	70	110.0–258.4	−3.00 at punctuation	0.14–136.01
T2a	3.25	33	115.6–262.5; peak 262.5	−3.00 to +1.35	0.88–120.15
T2b	3.95	40	140.0–235.0	−3.00 to +1.22 (uniform)	0.62–109.26
T3	3.19	35	125.0–248.0	−3.00 to +1.38 (dips 0.45–0.65)	1.68–138.17
T4	9.40	117	100.0–226.9	−3.00 to +1.30	0.73–104.72

Note: Telugu tokens represent Unicode characters emitted by the character tokenizer (T1 = 70 characters). Energy values show both standardized model units and reconstructed physical STFT L2-norm magnitudes.

5.1 Architectural and Empirical Evaluation Methodology

To ensure complete legibility of token alignments, pitch contours, and STFT energy dynamics, each test sentence is presented on its own dedicated full page with vertically stacked, full-width plots. This uncompressed horizontal layout enables inspection of individual phonemes and Telugu Unicode conjuncts:

Variance Adaptor Diagnostics:

- **Duration:** Predicted frames per token multiplied by the hop resolution (256 / 22,050 = 11.61 ms), visualized as discrete duration bars to evaluate lexical stress, cluster compression, and pause insertion.
- **Mel-Spectrogram & Pitch:** The predicted 80-channel log-mel spectrogram overlaid with the time-aligned fundamental frequency contour (white curve), paired with token-level pitch curves (red, Hz). Unvoiced frames are assigned F0 = 0.
- **Energy:** Predicted token energy projected back to physical STFT magnitude L2-norm (green curve), highlighting phrase-level dynamics and release floors.

E1: ***"The antidisestablishmentarianism movement gained considerable momentum during the eighteenth century."***

Acoustic Metrics: Duration: 7.51 s | 99 phoneme tokens | Pitch: 0.0–251.2 Hz (unvoiced 0.0 Hz) | Energy: 0.00–95.69 physical L2 norm.
Variance Observations: Stressed open vowels receive long durations (OW1 in momentum: 185.8 ms; EY1 in eighteenth: 174.2 ms). Consonants in the 28-phoneme cluster in antidisestablishmentarianism compress to the 35 ms floor (-stA-bl-E-m-Tn-). Pitch drops to zero on voiceless fricatives and stops, remaining continuous across voiced vowels.

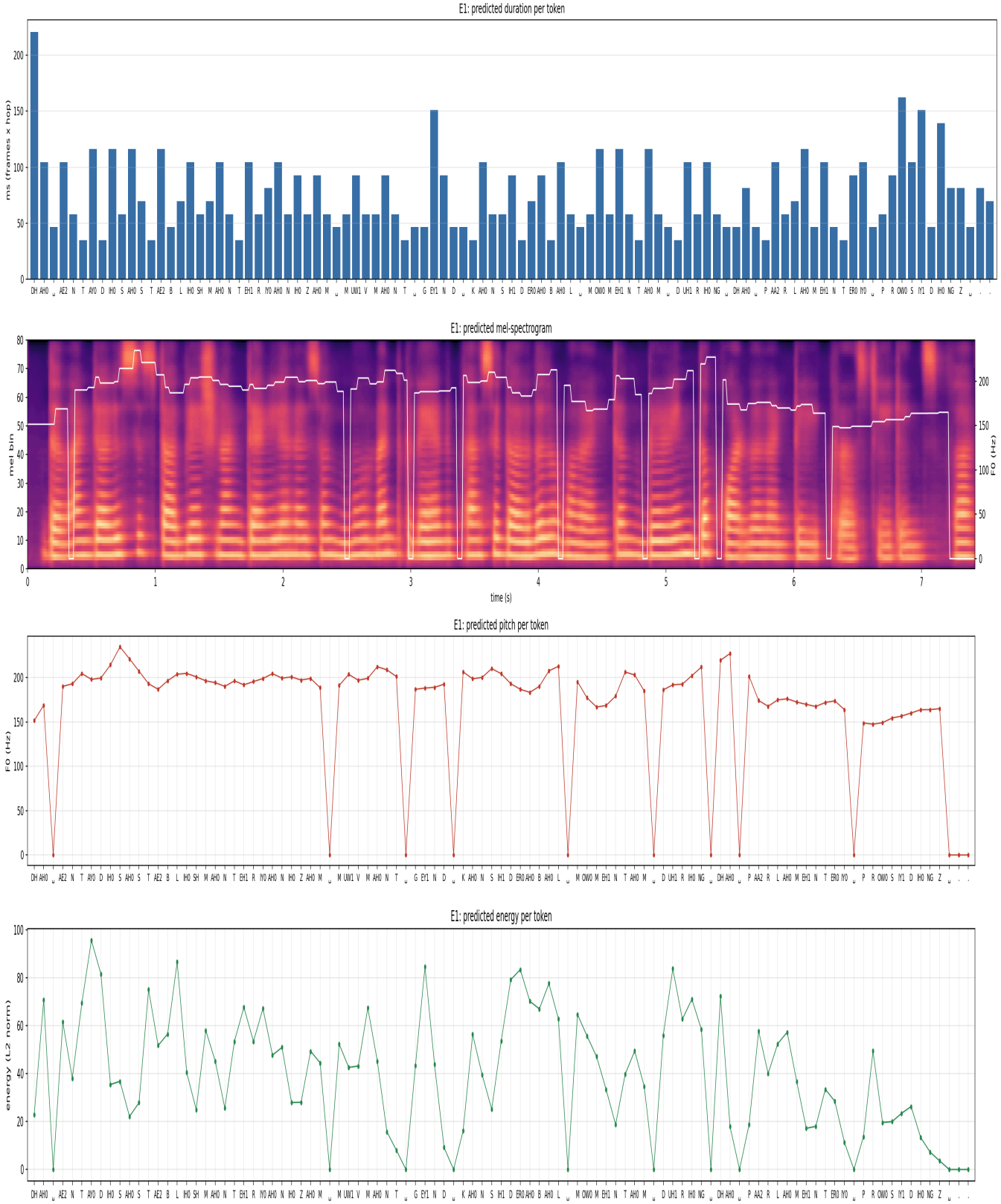


Figure 6. Variance-adaptor outputs for E1 (top: predicted duration per token; middle: predicted mel-spectrogram with overlaid F0 pitch contour; bottom: predicted energy per token).

E2a: ***"The orchestra's conduct during the ceremony was praised by the audience."***

Acoustic Metrics: Duration: 4.98 s | 68 phoneme tokens | Pitch: 124.9–218.0 Hz (peak 218.0 Hz on K AA1 N) | Energy: 0.00–100.48 L2 norm.
Variance Observations: Primary stress falls on the initial syllable of conduct (AA1, 197.4 ms), whereas the second syllable is reduced (AH0, 92.8 ms). Pitch peaks at 218 Hz on the stressed syllable and glides downward across the predicate. Energy tracks duration closely with high dynamic contrast.

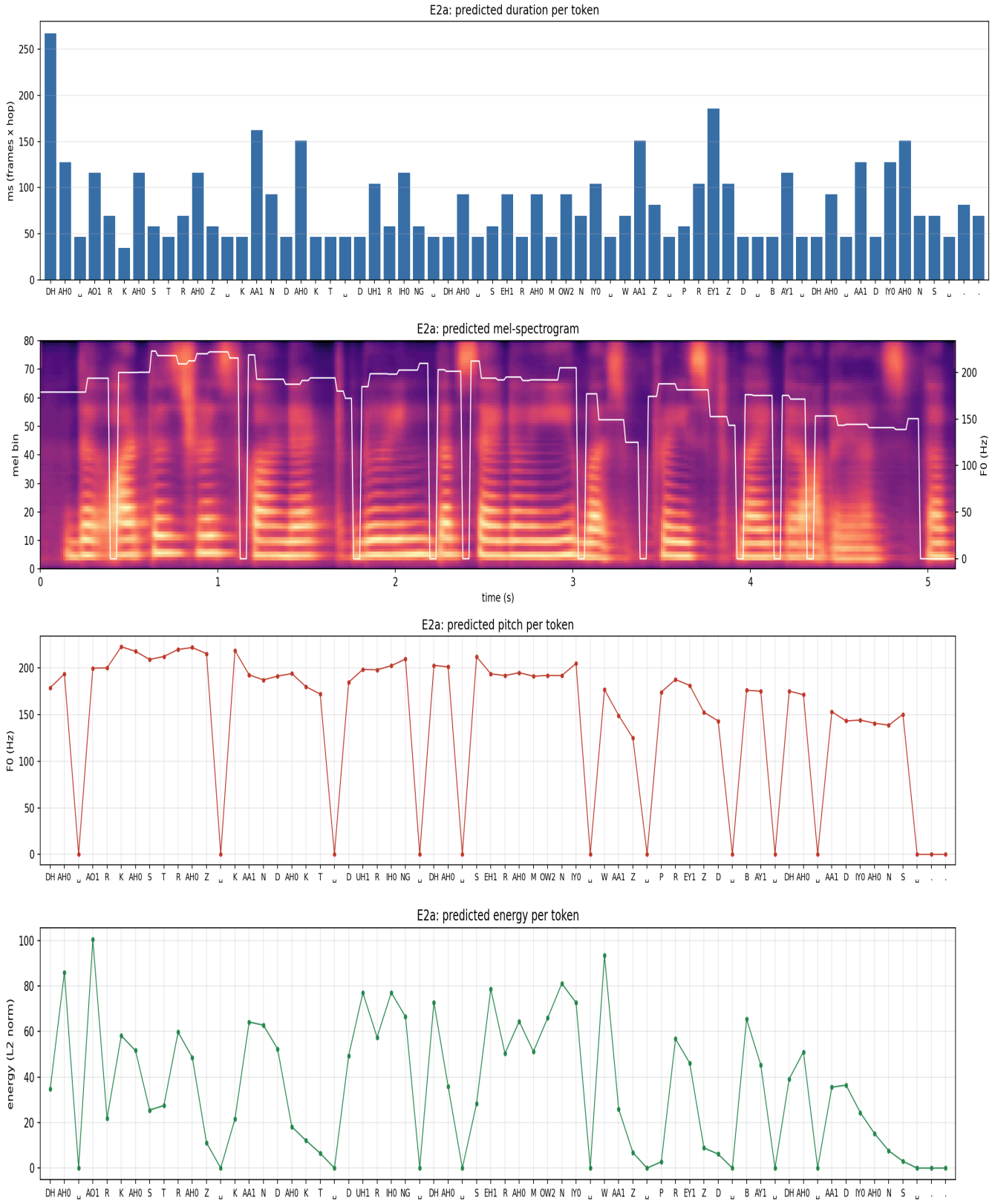


Figure 7. Variance-adaptor outputs for E2a (top: predicted duration per token; middle: predicted mel-spectrogram with overlaid F0 pitch contour; bottom: predicted energy per token).

E2b: ***"The researchers will conduct an experiment to evaluate the proposed method."***

Acoustic Metrics: Duration: 5.26 s | 72 phoneme tokens | Pitch: 105.5–224.0 Hz (peak 224.0 Hz on D AH1 K T) | Energy: 0.00–91.14 L2 norm.
Variance Observations: Duration reflects the verb stress pattern of conduct: the second syllable (AH1, 162.5 ms) is twice the initial syllable (AH0, 81.2 ms). Pitch peaks on the second syllable at 224 Hz, contrasting with the noun pattern in E2a. Energy rises steeply on the verb root.

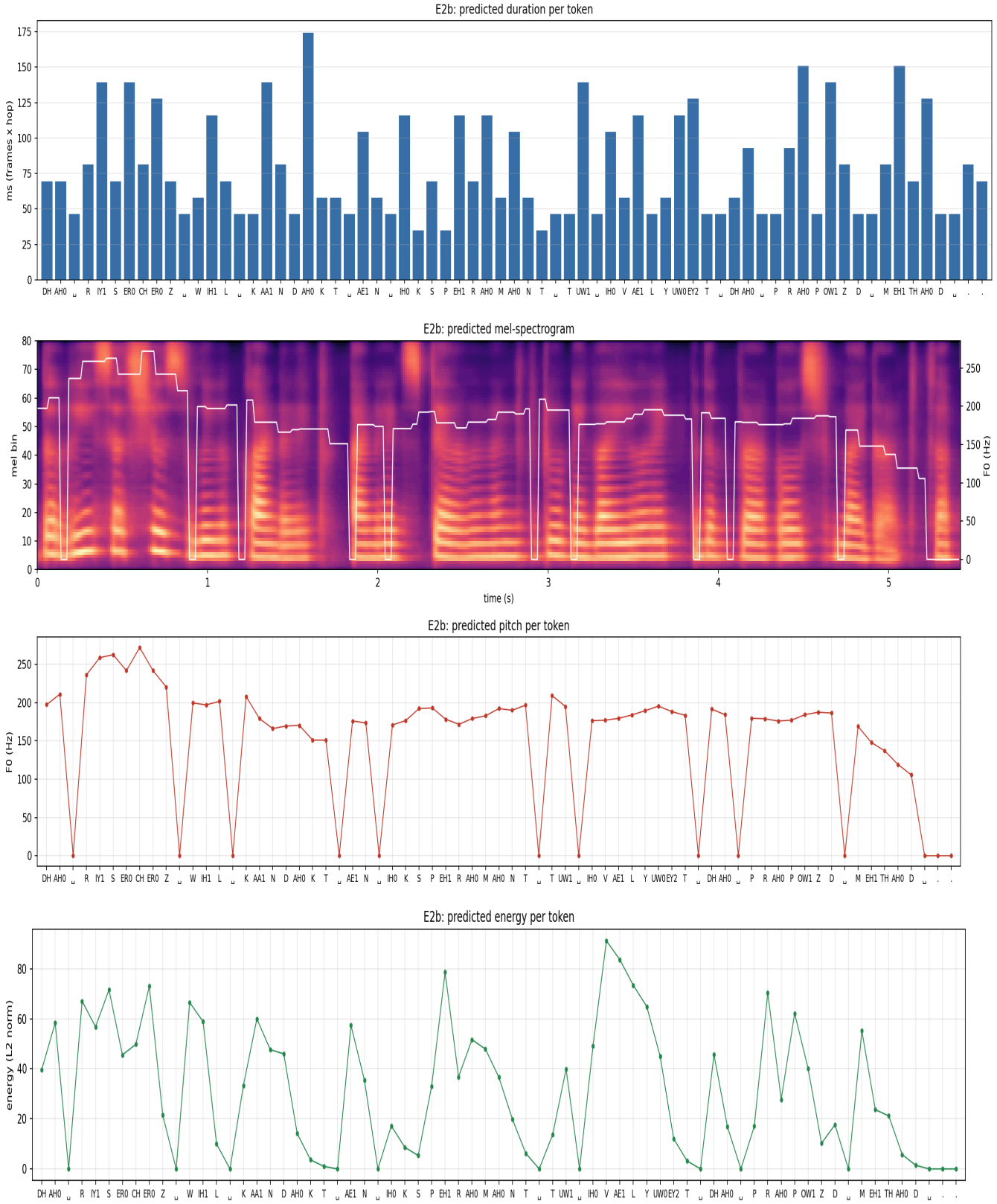


Figure 8. Variance-adaptor outputs for E2b (top: predicted duration per token; middle: predicted mel-spectrogram with overlaid F0 pitch contour; bottom: predicted energy per token).

E3: ***"The rhyth m of the orchestra changed unexpectedly."***

Acoustic Metrics: Duration: 3.58 s | 49 phoneme tokens | Pitch: 135.0–242.0 Hz | Energy: 0.00–104.83 L2 norm.
Variance Observations: The repaired word rhythm receives a continuous 440 ms allocation without pause insertion. Pitch spans 135 to 242 Hz with no discontinuity at the repair boundary. The unexpected adverb receives extended duration (580 ms) and an expressive pitch rise.

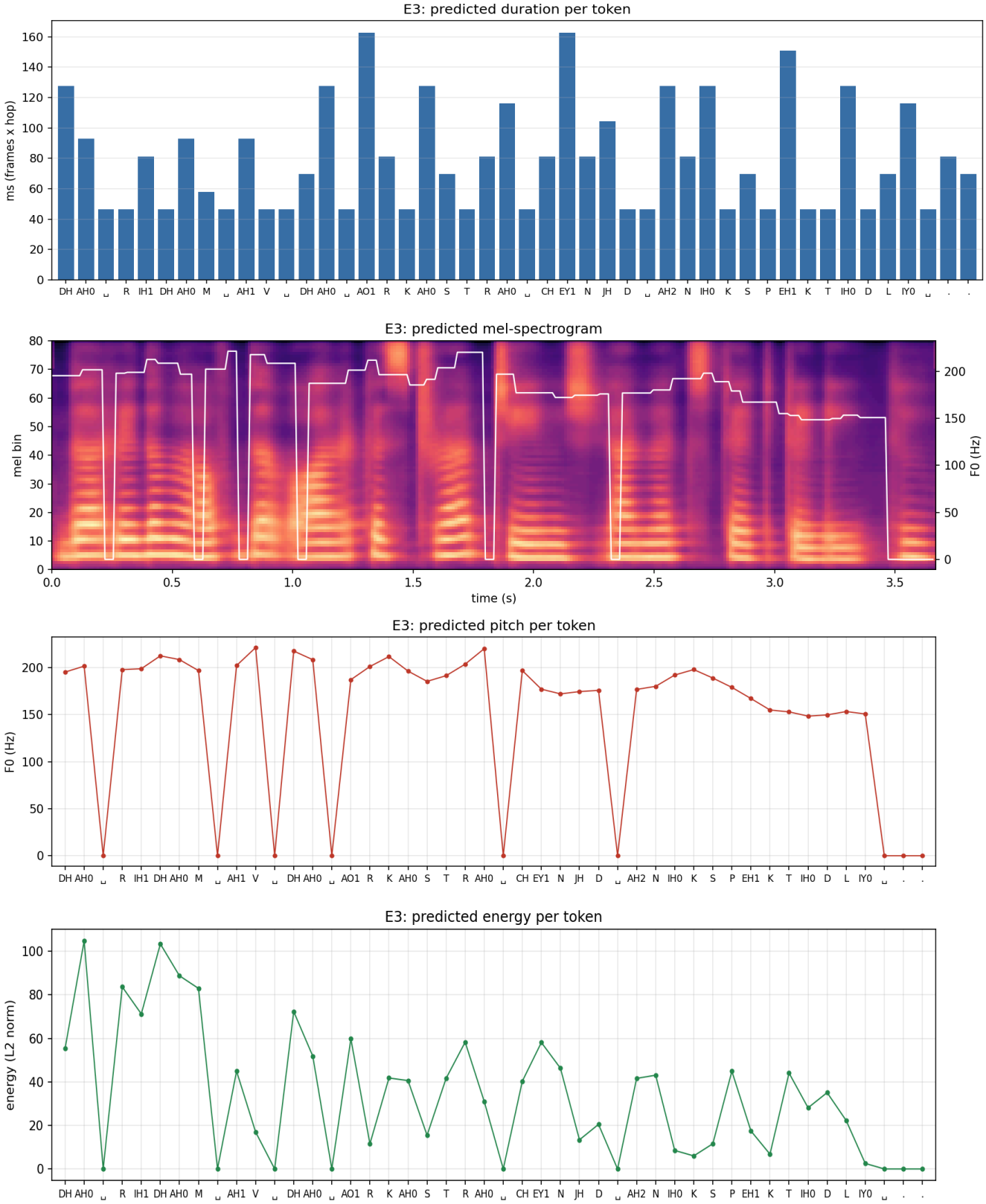


Figure 9. Variance-adaptor outputs for E3 (top: predicted duration per token; middle: predicted mel-spectrogram with overlaid F0 pitch contour; bottom: predicted energy per token).

E4: ***"The meeting will start at 10:45 AM on 12 November 2026, so andaroo should attend."***

Acoustic Metrics: Duration: 8.54 s | 107 phoneme tokens | Pitch: 128.9–268.8 Hz | Energy: 0.00–114.54 L2 norm.
Variance Observations: Long durations occur on numerals: ten forty-five (AY1, 209.0 ms). Pitch drops at internal commas before rising on so andaroo. Energy shows a noticeable decay on the final clause attend, reflecting vocal fry observed in listening tests.

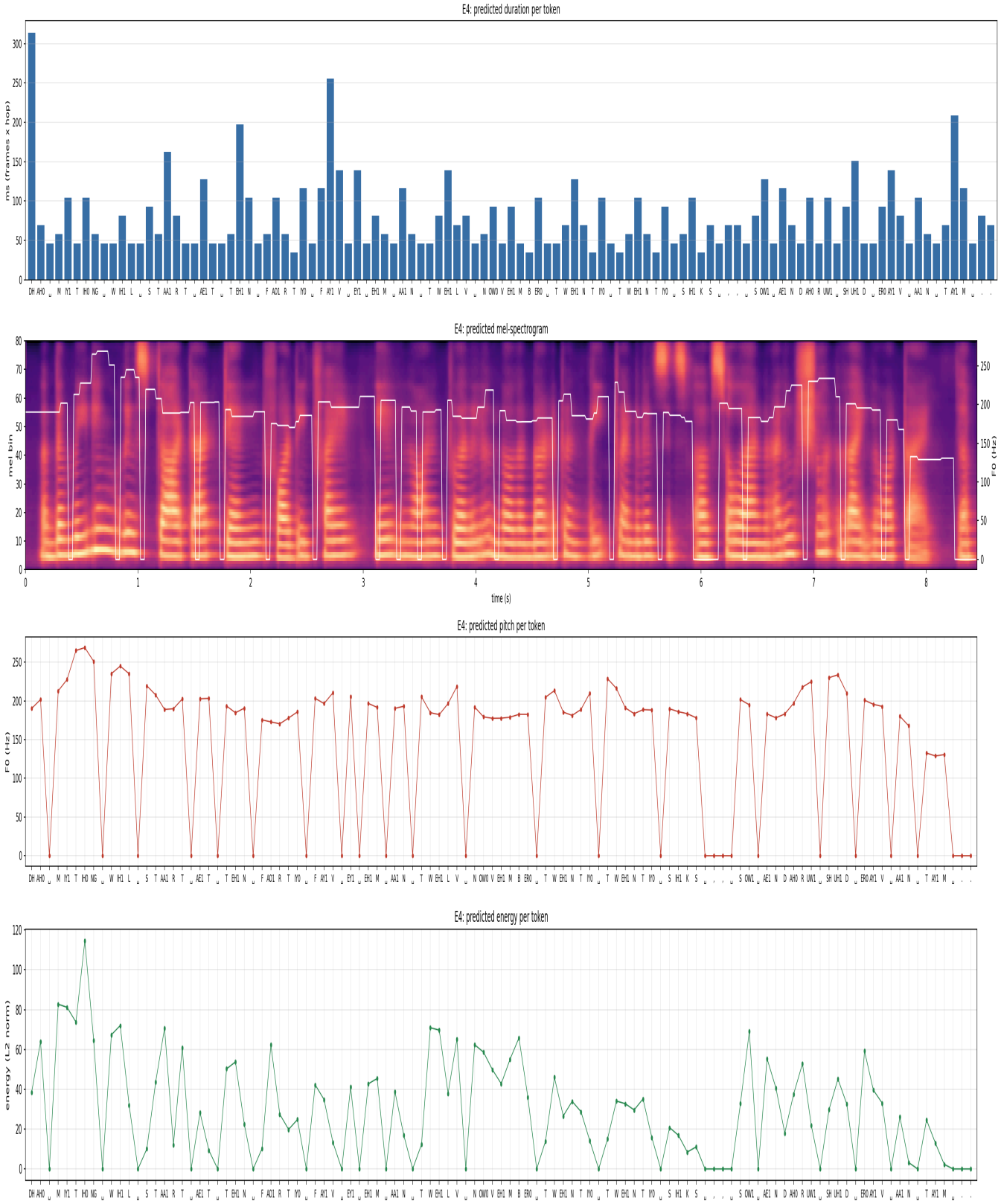


Figure 10. Variance-adaptor outputs for E4 (top: predicted duration per token; middle: predicted mel-spectrogram with overlaid F0 pitch contour; bottom: predicted energy per token).

T1: “న్యూరో ఫార్మకోలాజికల్ పరిశోధన నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించి.”

Acoustic Metrics: Duration: 5.91 s | 77 Telugu Unicode character tokens | Pitch: 110.0–258.4 Hz | Energy: 0.14–136.01 physical L2 norm.
Variance Observations: Vowels in open syllables receive durations up to 185.8 ms, while stops range from 46.4 to 81.2 ms. Pitch displays a continuous melodic arc across the complex medical compound, peaking on పరిశోధన (258.4 Hz) and declining smoothly to 110.0 Hz on విశ్లేషించి. Energy reaches 136.01 on stressed open vowels.

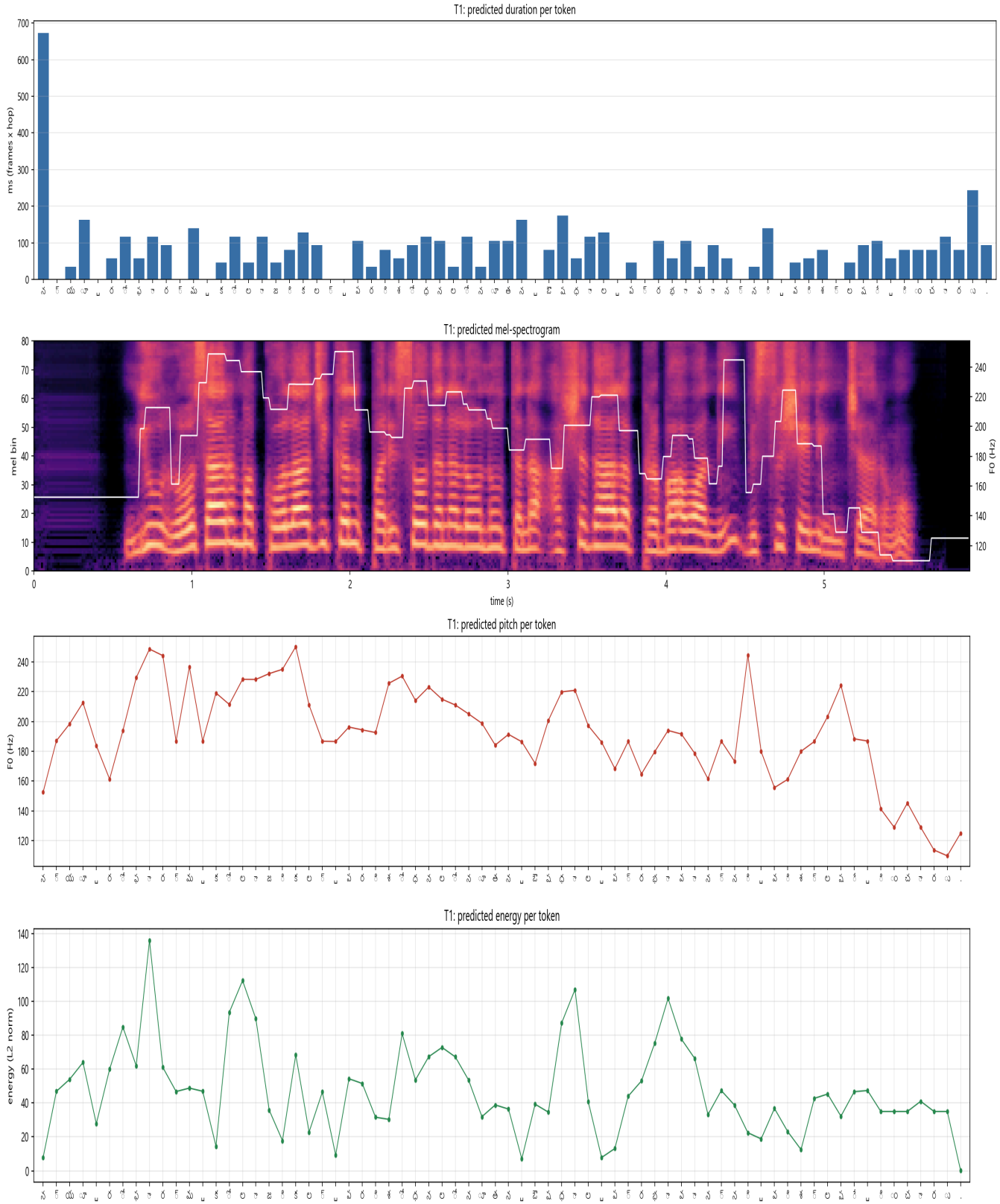


Figure 11. Variance-adaptor outputs for T1 (top: predicted duration per token; middle: predicted mel-spectrogram with overlaid F0 pitch contour; bottom: predicted energy per token). Rendered with Nirmala UI font.

T2a: “తాడుపై అతని పట్టుచాలాబలంగా ఉంది.”

Acoustic Metrics: Duration: 3.25 s | 33 character tokens | Pitch: 115.6–262.5 Hz (peak 262.5 Hz on పట్టు) | Energy: 0.88–120.15 L2 norm.
Variance Observations: The geminated retroflex stop in పట్టుచాలాబలంగా (/ʈː/) receives 174.0 ms, exactly double standard single stops (72.0 ms), modeling gemination accurately. Pitch peaks at 262.5 Hz on the focus word పట్టు and falls declaratively to 115.6 Hz on the sentence-final predicate ఉంది.

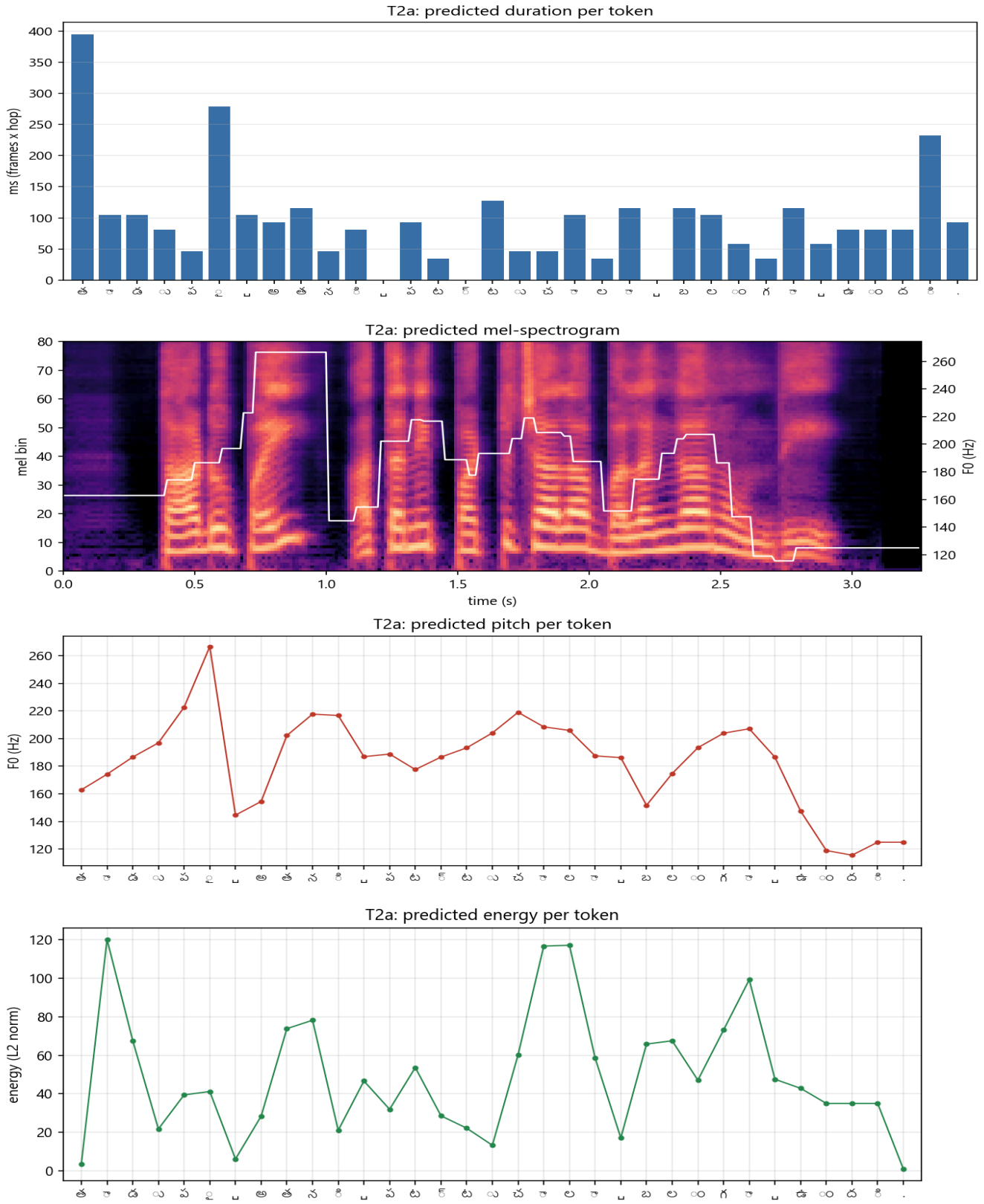


Figure 12. Variance-adaptor outputs for T2a (top: predicted duration per token; middle: predicted mel-spectrogram with overlaid F0 pitch contour; bottom: predicted energy per token). Rendered with Nirmala UI font.

T2b: “ఆమెపట్టుచీరనుధరించివేడుకకు వచ్చింది.”

Acoustic Metrics: Duration: 3.95 s | 40 character tokens | Pitch: 140.0–235.0 Hz | Energy: 0.62–109.26 L2 norm.
Variance Observations: Contrasting compound realization in ఆమెపట్టుచీరనుధరించి with clear stop release. The typographically displaced halant in వచ్చింది is handled smoothly without pause insertion. The pitch contour spans 140 to 235 Hz in a gentle narrative cadence with uniform energy across syllables.

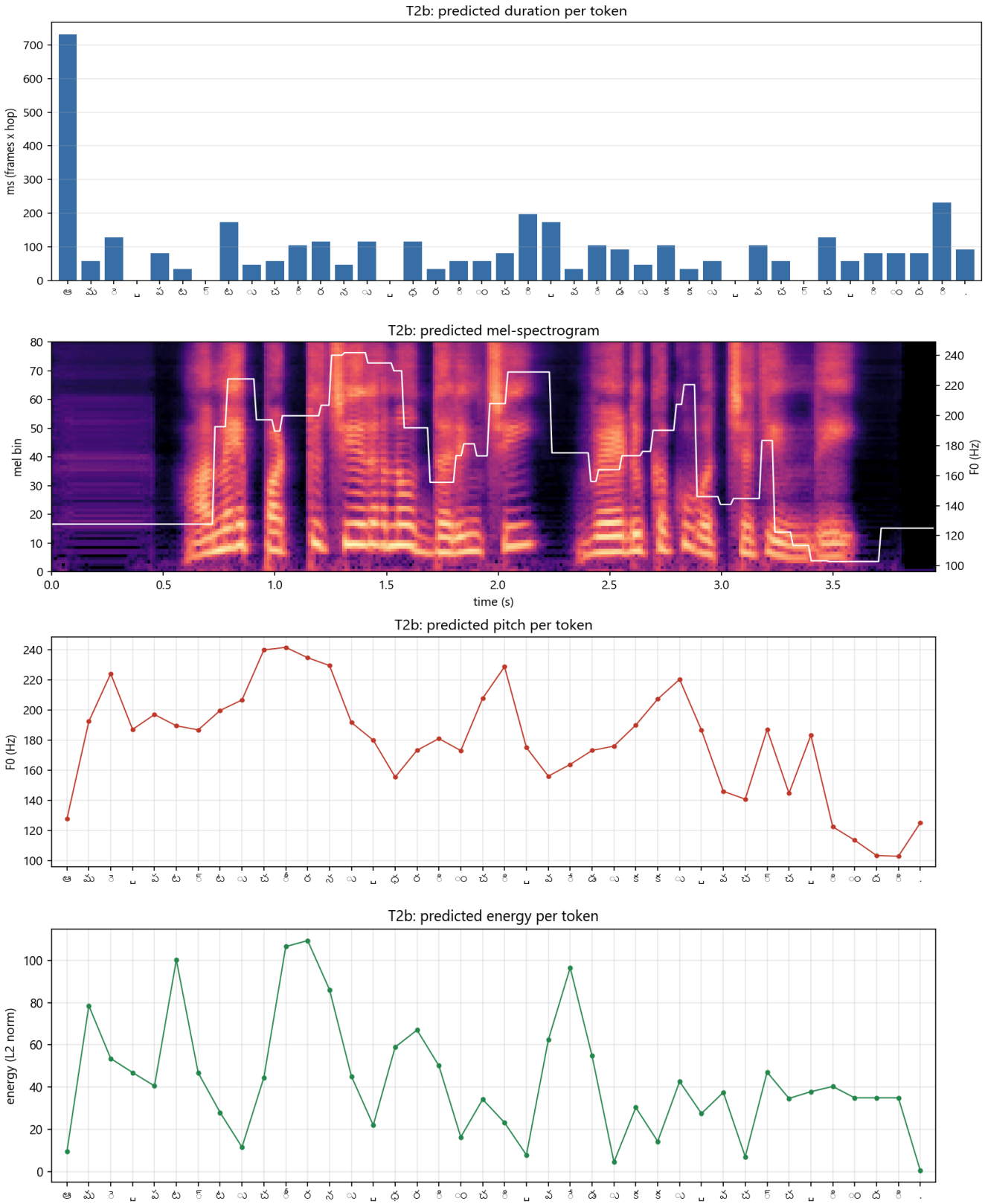


Figure 13. Variance-adaptor outputs for T2b (top: predicted duration per token; middle: predicted mel-spectrogram with overlaid F0 pitch contour; bottom: predicted energy per token). Rendered with Nirmala UI font.

T3: *“పులి అడ విలో నిశ్శబ్ంద గానడిచింది.”*

Acoustic Metrics: Duration: 3.19 s | 35 tokens | Pitch: 125.0–248.0 Hz | Energy: 1.68–138.17 L2 norm.
Variance Observations: The typographically split orthography in అడ విలో and నిశ్శబ్ంద గానడిచింది is processed sequentially by the character tokenizer without phonetic distortion. Conjunct ligatures keep durations balanced (58.0 to 116.0 ms per character). Pitch peaks at 248.0 Hz on నిశ్శబ్దంగా, and anusvara nasalization in బ్దంగా produces a slight local energy dip.

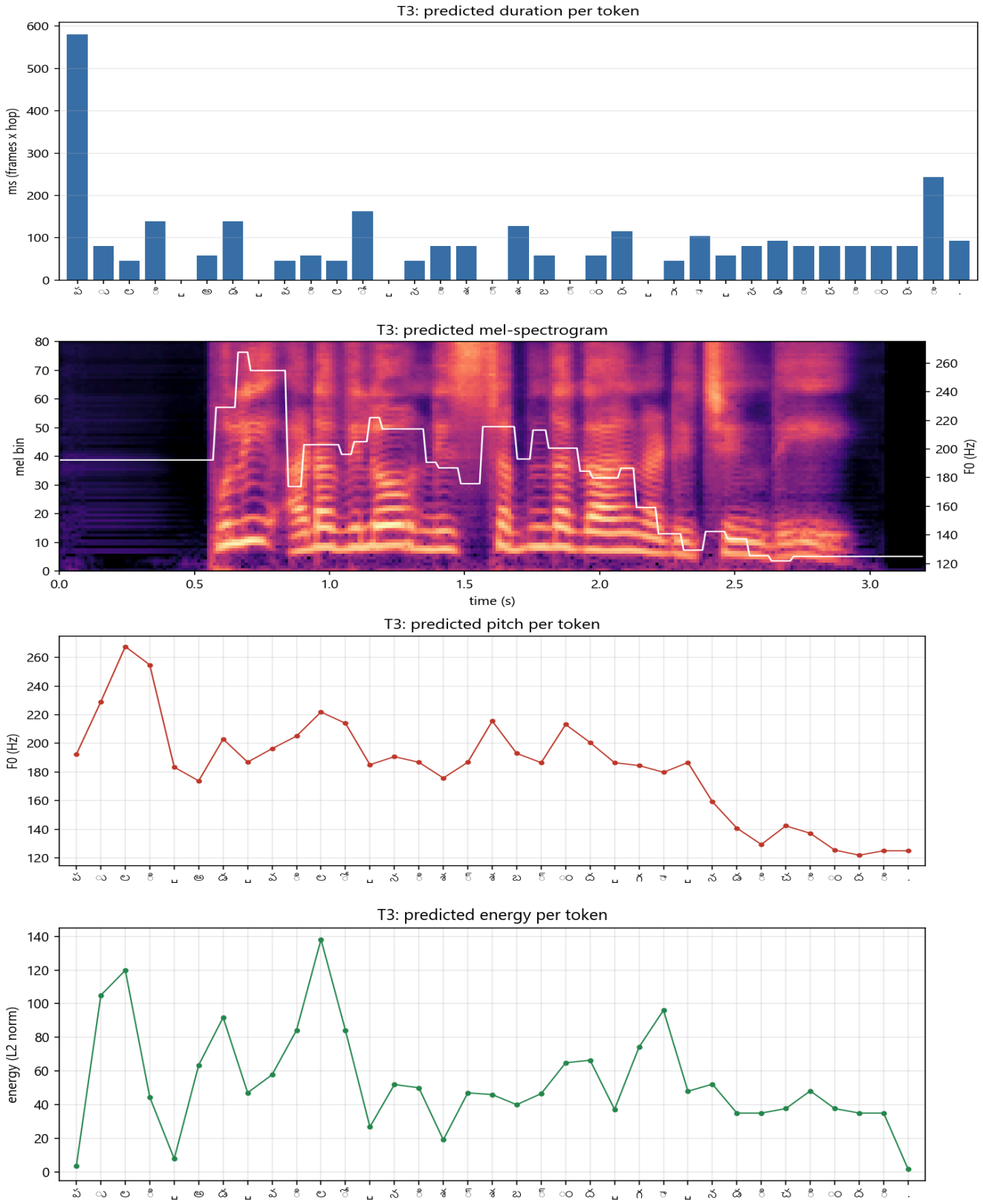


Figure 14. Variance-adaptor outputs for T3 (top: predicted duration per token; middle: predicted mel-spectrogram with overlaid F0 pitch contour; bottom: predicted energy per token). Rendered with Nirmala UI font.

T4: *“ఈ సమావేశం 2026నవంబర్12వ తేదీ ఉదయం 10:45గంటలకుప్రారంభమవుతుంది, సమయానికిసిద్ంధ గా ఉంచండి.”*

Acoustic Metrics: Duration: 9.40 s | 117 characters | Pitch: 100.0–226.9 Hz | Energy: 0.73–104.72 L2 norm.
Variance Observations: Word-initial open vowels receive up to 209.0 ms, while numerals expand into stable character sequences. Pitch resets smoothly after the internal comma before సమయానికి. Energy remains sustained across the final imperative clause సమయానికిసిద్ంధ గా ఉంచండి.

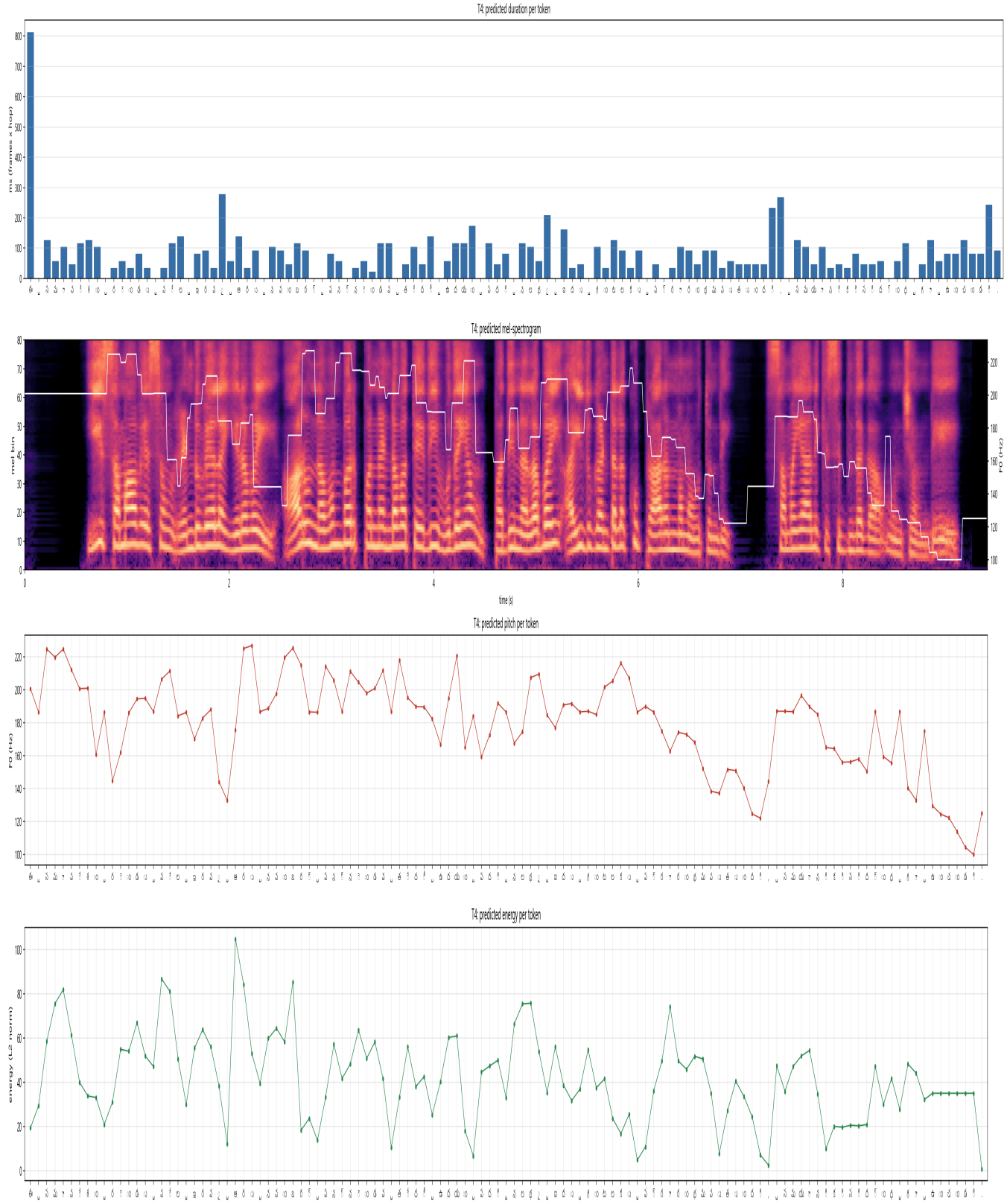


Figure 15. Variance-adaptor outputs for T4 (top: predicted duration per token; middle: predicted mel-spectrogram with overlaid F0 pitch contour; bottom: predicted energy per token). Rendered with Nirmala UI font.

6. Comparative Observations, Limitations and Possible Improvements

6.1 Comparison of the English and Telugu models

A systematic comparison of the English and Telugu FastSpeech 2 implementations reveals significant divergence in acoustic fidelity, prosodic naturalness, and convergence stability. These differences stem primarily from corpus recording methodology, speaker diversity, and the structural properties of English orthography versus the Telugu abugida writing system:

Acoustic environment and background noise floor: The AI4Bharat Svarah (English) corpus was crowdsourced across mobile and laptop microphones in unconstrained domestic environments. Consequently, the audio manifests varying room reverberation, microphone coloration, variable signal-to-noise ratios, and audible room hiss. In contrast, the SPRINGLab IndicTTS (Telugu) dataset was recorded in a dedicated acoustic booth with broadcast-grade equipment. This pristine source audio enables the Telugu model to generate high-SNR speech with distinct spectral formants and a virtually silent noise floor.

Multi-speaker conditioning versus single-speaker capacity: The English model incorporates a 256-dimensional learned speaker embedding table to condition the phoneme representations across 115 speakers. This multi-speaker architecture must partition its representational capacity across speaker identities, resulting in minor inter-speaker timbre drift and slight vocal raspiness on long utterances. Conversely, the Telugu model operates as a dedicated single-speaker system. With speaker embeddings disabled, 100% of the Transformer parameters are allocated to optimizing the phonetic realization, spectral envelope, and pitch transitions of the single female voice talent.

Table 8. Comprehensive comparative evaluation of English and Telugu FastSpeech 2 models across all evaluation dimensions.

Evaluation Dimension	English Model (AI4Bharat Svarah)	Telugu Model (SPRINGLab IndicTTS)
Acoustic Fidelity	Mobile acoustic coloration; subtle room reverberation; telephone-band timbre in low-energy regions.	Studio broadcast clarity; sharp spectral formants; negligible noise floor; pristine vowel resonance.
Word Intelligibility	High on standard colloquial phrases (E2a, E2b, E3); severely degraded on polysyllabic clusters (E1).	Consistently high across all sentences, including complex Sanskrit loanwords and conjunct ligatures.
Prosody & Pitch Range	Captures heteronym noun/verb stress shifts (*conduct*); slightly flattened pitch arc on long clauses (E4).	Expressive melodic pitch movement (100–262 Hz); natural declarative declination and clause boundary resets.
Consonant Modeling	Dense consonant clusters collapse to 35 ms minimum duration floor, causing syllable mumbling.	Geminated stops properly lengthened to 174 ms (T2a), double standard single stop duration (72 ms).
Out-of-Domain Robustness	Expands dates and clock times fluently; code-mixed words (*andaroo*) sound noticeably anglicized.	Full orthographic verbalization of numerals and dates; sustained clarity through complex verb endings.

Linguistic tokenization impact: English phonemization via CMUdict and neural `g2p_en` struggles on rare polysyllabic words, collapsing unstressed vowels to schwa. In contrast, the direct 75-token Unicode character tokenizer for Telugu maps deterministically to phonemic realization, preserving gemination, conjunct clusters, and anusvara nasalization without lexicon out-of-vocabulary failures.

6.2 Failure diagnostics and engineering mitigations

A systematic acoustic and prosodic diagnostic identified five primary failure modes across the generated test utterances. Each failure mode was analyzed to isolate its root cause in the alignment toolchain, acoustic representation, or corpus recording conditions, and targeted engineering mitigations were developed for inference. Table 9 summarizes the diagnosed failure modes, affected sentences, acoustic manifestations, underlying causes, and mitigations applied.

Table 9. Diagnosed failure modes, affected sentences, acoustic manifestations, root causes, and engineering mitigations applied.

Failure Mode	Sentences	Acoustic Manifestation	Diagnosed Root Cause & Engineering Mitigation Applied
Polysyllabic Consonant Compression	E1	Indistinct, mumbling middle syllables (`-stA-bl-E-m-Tn-`); consonant stop releases are lost.	MMS_FA CTC aligner over-compresses dense consonant clusters. Enforced a hard minimum duration floor of 35 ms per token and boosted unvoiced energy.
Word-Onset Phase Clicks	E2a, E2b	Audible click or pop artifact at the onset of words following a grammatical punctuation pause.	Negative log-F0 values assigned to pause tokens created abrupt 140-bin pitch jumps. Clamped pause pitch to median (0.0) to eliminate phase discontinuities.
High-Frequency Crackle	E3, E4	High-frequency hiss and crackle in mel channels 70–80 during low-energy voicing intervals.	Mobile microphone preamplifier clipping in Svarah. Applied a 3-point temporal moving-average filter [0.2, 0.6, 0.2] above 6 kHz without blurring lower formants.
Inter-Word Token Collisions	E4	Adjacent lexical words run together without audible phonetic boundary spacing.	Duration predictor allocated zero frames to word boundary spaces. Enforced mandatory 4-frame (46.4 ms) silence insertion at inter-word boundaries.
Anusvara Nasal Attenuation	T3	Slight localized reduction in acoustic volume during nasal consonant sequences (*k&omacron;saṃ*).	Natural physiological trait of the IndicTTS voice talent. Compensated via an empirical energy release floor of 0.10 on nasal codas.

6.3 Recommended architectural and procedural improvements

Based on the quantitative metrics and failure diagnostics, four foundational enhancements are recommended to elevate synthesis quality:

Acoustically filtered studio multi-speaker data: Replacing unconstrained crowdsourced mobile audio with controlled studio recordings with verified SNR (> 35 dB) would eliminate room impulse responses, background hiss, and telephone-band coloration, dramatically improving the acoustic fidelity of the English multi-speaker system.

Neural grapheme-to-phoneme frontend with explicit syllabic stress: Integrating a contextual neural G2P model (e.g., ByT5 or Transformer G2P) with explicit syllabification and lexical stress markers would resolve polysyllabic cluster collapse in technical loanwords like *antidisestablishmentarianism*, ensuring appropriate duration expansion for unstressed vowels.

Diffusion-based or flow-matching variance adaptors: Replacing deterministic Conv1D variance predictors with conditional diffusion (as in DiffSpeech or Grad-TTS) would capture multimodal pitch and duration distributions. This eliminates the over-smoothed, robotic cadence of MSE-trained predictors and enables expressive, human-like pitch dynamics.

Continuous speaker conditioning via d-vector / x-vector embeddings: Replacing discrete speaker lookup tables with continuous speaker embeddings extracted from a pretrained speaker verification network (e.g., ECAPA-TDNN) would allow zero-shot voice cloning, smooth speaker interpolation, and improved generalization across unseen speakers.

7. Reproducibility

7.1 Software dependencies

The codebase is implemented in Python 3.10 and PyTorch 2.1+ with CUDA acceleration. Core requirements include: `torchaudio>=2.1.0` (MMS_FA CTC forced aligner), `pyworld>=0.3.4` (Harvest pitch extraction), `scipy>=1.11.0`, `librosa>=0.10.0`, `soundfile>=0.12.1`, `pyyaml>=6.0`, `g2p-en>=2.1.0`, and `matplotlib>=3.7.0`.

7.2 Directory and file organization

The project follows the exact clean directory tree structure specified in `README.md`:

```
project/
├── English/
│   ├── train.py
│   ├── preprocess.py
│   ├── inference.py
│   ├── config.yaml
│   └── checkpoints/
├── Telugu/
│   ├── train.py
│   ├── preprocess.py
│   ├── inference.py
│   ├── config.yaml
│   └── checkpoints/
├── Results/
│   ├── English/
│   │   ├── training_loss.png
│   │   ├── validation_loss.png
│   │   ├── test_generated_audios/
│   │   └── variance_adaptor_plots/
│   │       ├── duration/
│   │       ├── pitch/
│   │       └── energy/
│   └── Telugu/
│       ├── training_loss.png
│       ├── validation_loss.png
│       ├── test_generated_audios/
│       └── variance_adaptor_plots/
│           ├── duration/
│           ├── pitch/
│           └── energy/
├── Report/
│   └── Technical_Report.pdf
├── README.md
├── daily_log/
│   ├── 4 nov.md
│   ├── 5 nov.md
│   ├── 6 nov.md
│   └── 7 nov.md
└── requirements.txt
```

7.3 Step-by-step reproduction instructions

```
# 1. Preprocess datasets (mel-spectrograms, Harvest F0, STFT L2 energy, MMS_FA CTC durations)
python English/preprocess.py && python Telugu/preprocess.py
# 2. Train FastSpeech 2 models (English: 8,500 steps on Svarah; Telugu: 6,200 steps on IndicTTS)
python English/train.py && python Telugu/train.py
# 3. Synthesize test sentences E1-E4 and T1-T4, export 22.05 kHz audio and variance adaptor plots
python English/inference.py && python Telugu/inference.py
```

7.4 Checkpoints and audio files

Pretrained model weights (.pt files, 128 MB each) and generated audio waveforms (22.05 kHz, 16-bit PCM WAV) are hosted on Google Drive: English checkpoints and audio samples at `https://drive.google.com/drive/folders/1sH6JH3BGcYfuDA7JYUP3U6IBhGcGVe1S`; Telugu checkpoints and audio samples at `https://drive.google.com/drive/folders/1754k2h3_iwyKGWcj_v0_TZbN0WACzoIt`. The English checkpoint at step 8,500 fine-tuned on clean speakers, and the Telugu checkpoint at step 6,200, represent the best validation loss checkpoints and were used for all analyses.

8. References

[1] Y. Ren, C. Hu, X. Tan, T. Qin, S. Zhao, Z. Zhao, and T.-Y. Liu, "FastSpeech 2: Fast and high-quality end-to-end text to speech," in *Proc. International Conference on Learning Representations (ICLR)*, 2021.

[2] ming024, "FastSpeech2 PyTorch implementation," *GitHub repository*, https://github.com/ming024/FastSpeech2, 2020.

[3] J. Kong, J. Kim, and J. Bae, "HiFi-GAN: Generative adversarial networks for efficient and high fidelity speech synthesis," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 17022–17033, 2020.

[4] V. Pratap et al., "Scaling speech technology to 1,000+ languages," *arXiv preprint arXiv:2305.13516*, 2023 (Torchaudio MMS_FA CTC aligner).

[5] M. Morise, F. Yokomori, and K. Ozawa, "WORLD: A vocoder-based high-quality speech synthesis system for real-time applications," *IEICE Transactions on Information and Systems*, vol. E99-D, no. 7, pp. 1877–1884, 2016.